



# Advanced Linear Algebra: Foundations to Frontiers

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## 3.2.1 Classical Gram-Schmidt (CGS) ¶

### 03.2.1 Gram Schmidt orthogonalization, part 1

fit width



Given a set of linearly independent vectors  $\{a_0, \dots, a_{n-1}\} \subset \mathbb{C}^m$ , the Gram-Schmidt process computes an orthonormal basis  $\{q_0, \dots, q_{n-1}\}$  that spans the same subspace as the original vectors, i.e.

$$\text{Span}(\{a_0, \dots, a_{n-1}\}) = \text{Span}(\{q_0, \dots, q_{n-1}\}).$$

The process proceeds as follows:

- Compute vector  $q_0$  of unit length so that  $\text{Span}(\{a_0\}) = \text{Span}(\{q_0\})$ :

- $\rho_{0,0} = \|a_0\|_2$

Computes the length of vector  $a_0$ .

- $q_0 = a_0 / \rho_{0,0}$

Sets  $q_0$  to a unit vector in the direction of  $a_0$ .

Notice that  $a_0 = q_0 \rho_{0,0}$

- Compute vector  $q_1$  of unit length so that  $\text{Span}(\{a_0, a_1\}) = \text{Span}(\{q_0, q_1\})$ :

- $\rho_{0,1} = q_0^H a_1$

Computes  $\rho_{0,1}$  so that  $\rho_{0,1} q_0 = q_0^H a_1 q_0$  equals the component of  $a_1$  in the direction of  $q_0$ .

- $a_1^\perp = a_1 - \rho_{0,1}q_0$

Computes the component of  $a_1$  that is orthogonal to  $q_0$ .

- $\rho_{1,1} = \|a_1^\perp\|_2$

Computes the length of vector  $a_1^\perp$ .

- $q_1 = a_1^\perp / \rho_{1,1}$

Sets  $q_1$  to a unit vector in the direction of  $a_1^\perp$ .

Notice that

$$\left( \begin{array}{c|c} a_0 & a_1 \end{array} \right) = \left( \begin{array}{c|c} q_0 & q_1 \end{array} \right) \left( \begin{array}{c|c} \rho_{0,0} & \rho_{0,1} \\ \hline 0 & \rho_{1,1} \end{array} \right).$$

- Compute vector  $q_2$  of unit length so that

$\text{Span}(\{a_0, a_1, a_2\}) = \text{Span}(\{q_0, q_1, q_2\})$ :

- $\rho_{0,2} = q_0^H a_2$  or, equivalently,  $\begin{pmatrix} \rho_{0,2} \\ \rho_{1,2} \end{pmatrix} = \begin{pmatrix} q_0 & q_1 \end{pmatrix}^H a_2$

Computes  $\rho_{0,2}$  so that  $\rho_{0,2}q_0 = q_0^H a_2 q_0$  and  $\rho_{1,2}q_1 = q_1^H a_2 q_1$  equal the components of  $a_2$  in the directions of  $q_0$  and  $q_1$ .

Or, equivalently,  $\begin{pmatrix} q_0 & q_1 \end{pmatrix} \begin{pmatrix} \rho_{0,2} \\ \rho_{1,2} \end{pmatrix}$  is the component in  $\text{Span}(\{q_0, q_1\})$ .

- $a_2^\perp = a_2 - \rho_{0,2}q_0 - \rho_{1,2}q_1 = a_2 - \begin{pmatrix} q_0 & q_1 \end{pmatrix} \begin{pmatrix} \rho_{0,2} \\ \rho_{1,2} \end{pmatrix}$

Computes the component of  $a_2$  that is orthogonal to  $q_0$  and  $q_1$ .

- $\rho_{2,2} = \|a_2^\perp\|_2$

Computes the length of vector  $a_2^\perp$ .

- $q_2 = a_2^\perp / \rho_{2,2}$

Sets  $q_2$  to a unit vector in the direction of  $a_2^\perp$ .

Notice that

$$\left( \begin{array}{cc|c} a_0 & a_1 & a_2 \end{array} \right) = \left( \begin{array}{cc|c} q_0 & q_1 & q_2 \end{array} \right) \left( \begin{array}{cc|c} \rho_{0,0} & \rho_{0,1} & \rho_{0,2} \\ \hline 0 & \rho_{1,1} & \rho_{1,2} \\ \hline 0 & 0 & \rho_{2,2} \end{array} \right).$$

- And so forth.

### 03.2.1 Gram Schmidt orthogonalization, part 2

fit width



Yet another way of looking at this problem is as follows.

### 03.2.1 Gram Schmidt orthogonalization, part 3

fit width



Consider the matrices

$$A = \left( \begin{array}{c|c|c|c|c|c|c} a_0 & \cdots & a_{k-1} & a_k & a_{k+1} & \cdots & a_{n-1} \end{array} \right)$$

and

$$Q = \left( \begin{array}{c|c|c|c|c|c|c} q_0 & \cdots & q_{k-1} & q_k & q_{k+1} & \cdots & q_{n-1} \end{array} \right)$$

We observe that

- $\text{Span}(\{a_0\}) = \text{Span}(\{q_0\})$

Hence  $a_0 = \rho_{0,0}q_0$  for some scalar  $\rho_{0,0}$ .

- $\text{Span}(\{a_0, a_1\}) = \text{Span}(\{q_0, q_1\})$

Hence

$$a_1 = \rho_{0,1}q_0 + \rho_{1,1}q_1$$

for some scalars  $\rho_{0,1}, \rho_{1,1}$ .

- In general,  $\text{Span}(\{a_0, \dots, a_{k-1}, a_k\}) = \text{Span}(\{q_0, \dots, q_{k-1}, q_k\})$

Hence

$$a_k = \rho_{0,k}q_0 + \dots + \rho_{k-1,k}q_{k-1} + \rho_{k,k}q_k$$

for some scalars  $\rho_{0,k}, \dots, \rho_{k,k}$ .

Let's assume that  $q_0, \dots, q_{k-1}$  have already been computed and are mutually orthonormal. Consider

$$a_k = \rho_{0,k}q_0 + \dots + \rho_{k-1,k}q_{k-1} + \rho_{k,k}q_k.$$

Notice that

$$\begin{aligned} q_k^H a_k &= q_k^H (\rho_{0,k}q_0 + \dots + \rho_{k-1,k}q_{k-1} + \rho_{k,k}q_k) \\ &= \rho_{0,k} \underbrace{q_k^H q_0}_0 + \dots + \rho_{k-1,k} \underbrace{q_k^H q_{k-1}}_0 + \rho_{k,k} \underbrace{q_k^H q_k}_1 \end{aligned}$$

so that

$$\rho_{i,k} = q_i^H a_k,$$

for  $i = 0, \dots, k-1$ . Next, we can compute

$$a_k^\perp = a_k - \rho_{0,k}q_0 - \dots - \rho_{k-1,k}q_{k-1}$$

and, since  $\rho_{k,k}q_k = a_k^\perp$ , we can choose

$$\rho_{k,k} = \|a_k^\perp\|_2$$

and

$$q_k = a_k^\perp / \rho_{k,k}$$

**Remark 3.2.1.1.** For a review of Gram-Schmidt orthogonalization and exercises orthogonalizing real-valued vectors, you may want to look at Linear Algebra: Foundations to Frontiers (LAFF) [27] Week 11.